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(54) **CONTAINER SYSTEM AND METHOD**

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(57)

ABSTRACT

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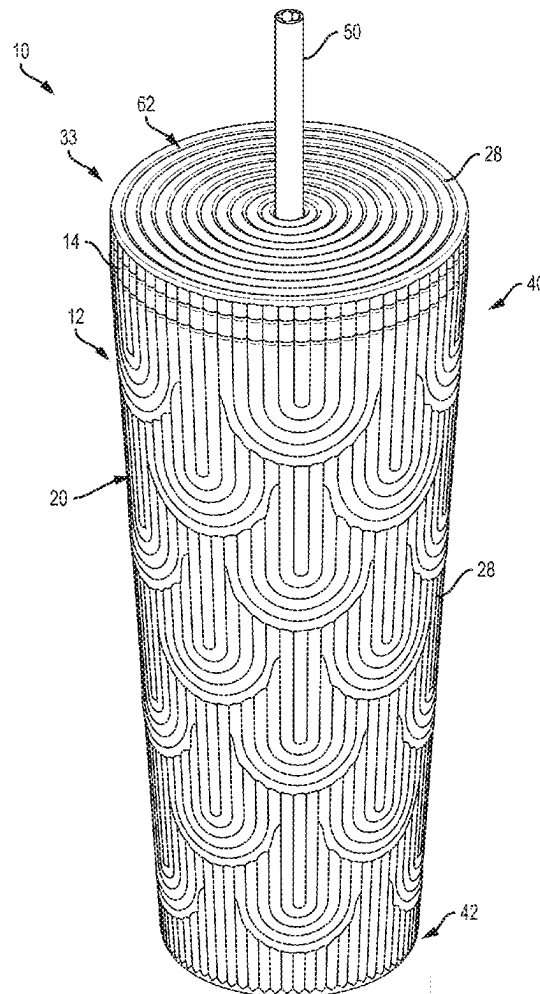
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A container system and a method of forming such a container system includes one or more lenticular lenses on an exterior surface of the container system. The disclosed container systems provide a visual characteristic that may appear different to the viewer/user when the container system is moved or viewed from a different angle. For example, when rotated or tilted with respect to the viewer, a container system that appears to be a first color originally may then appear to be a second color. In another example, when rotated or tilted with respect to the viewer, a container system that appears to have a first design. The color(s) and/or design(s) may be provided via an insert that fits between the exterior layer of the container system and an interior layer of the container system.



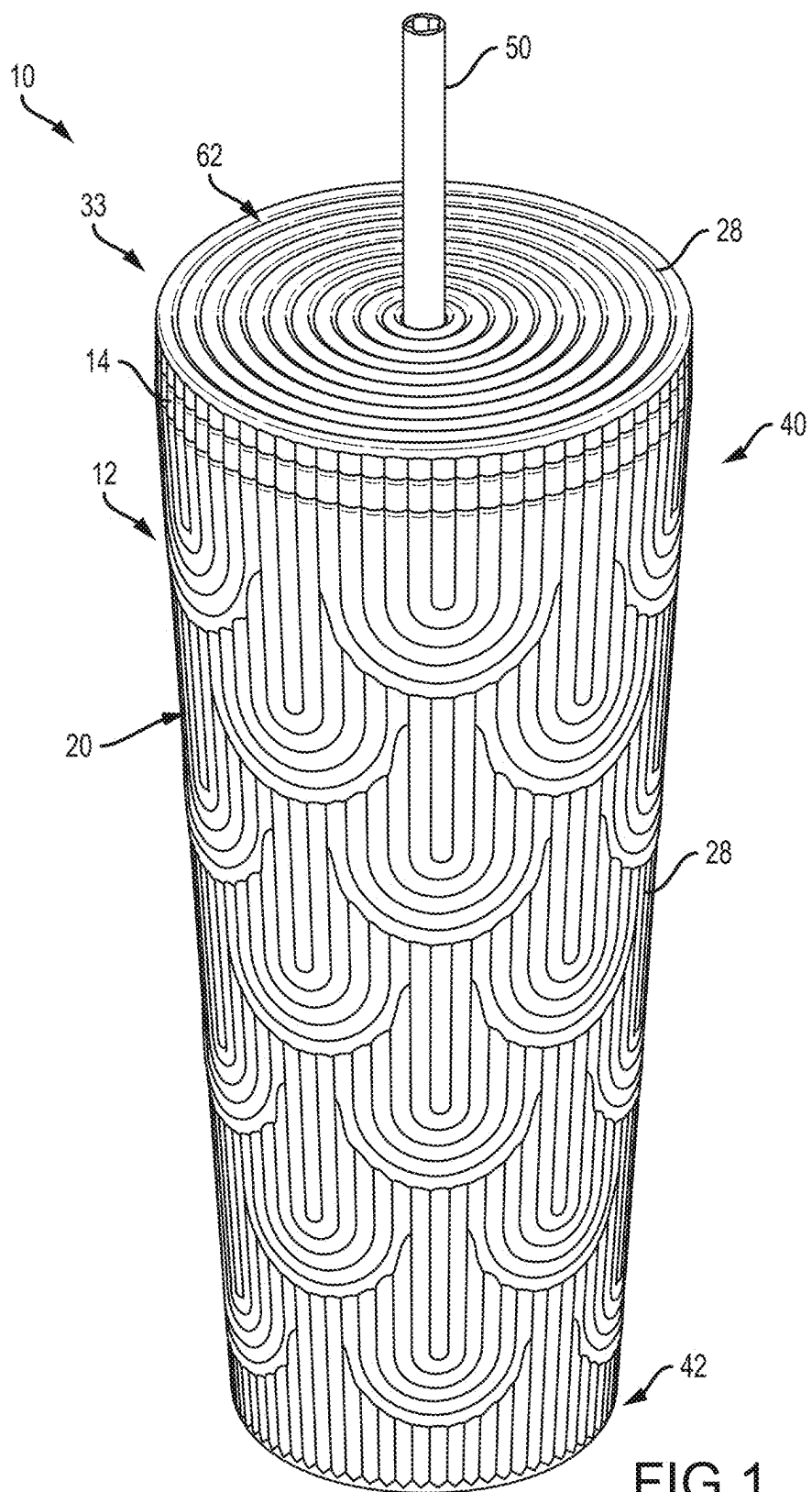


FIG.1

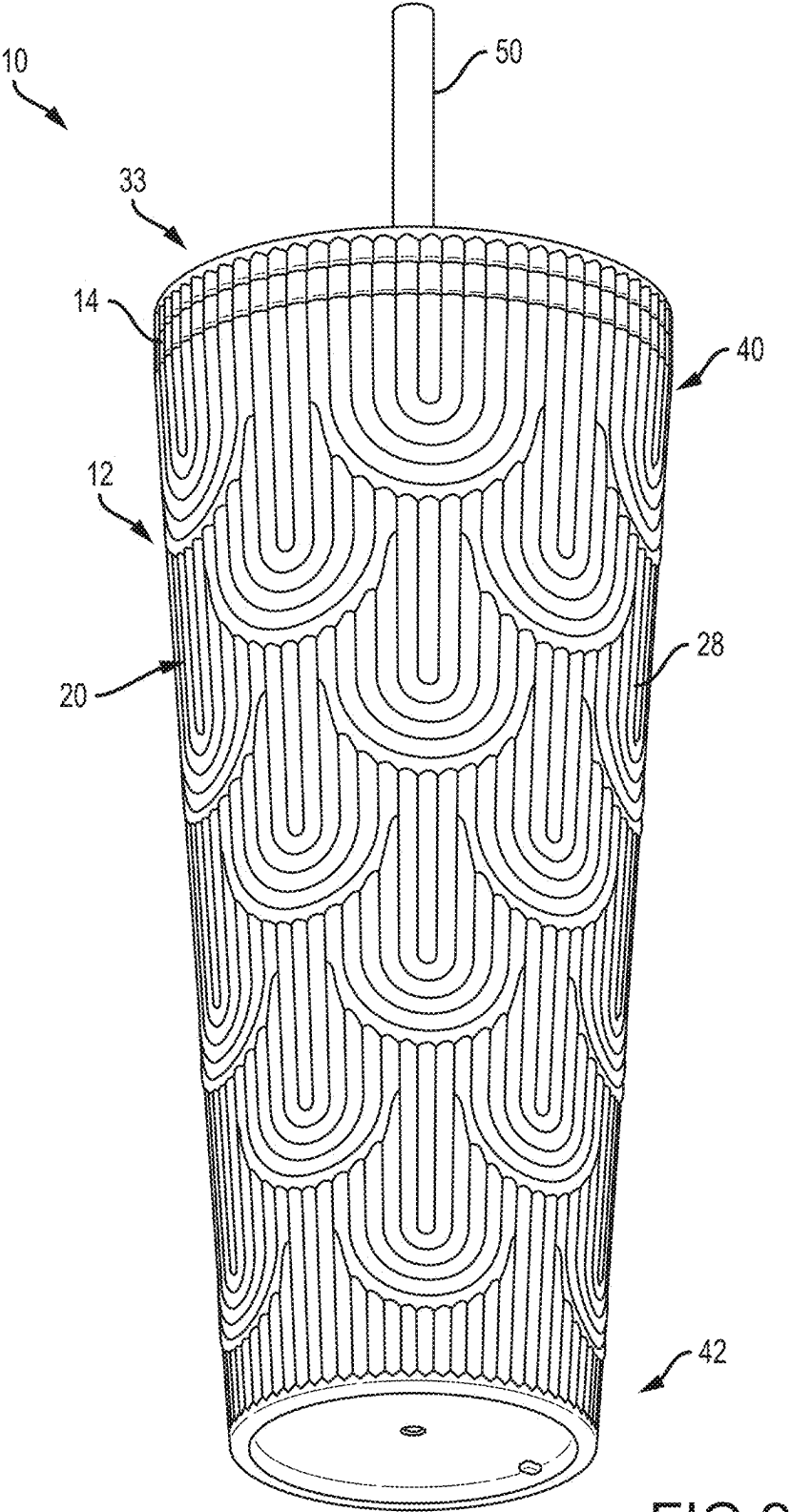


FIG.2

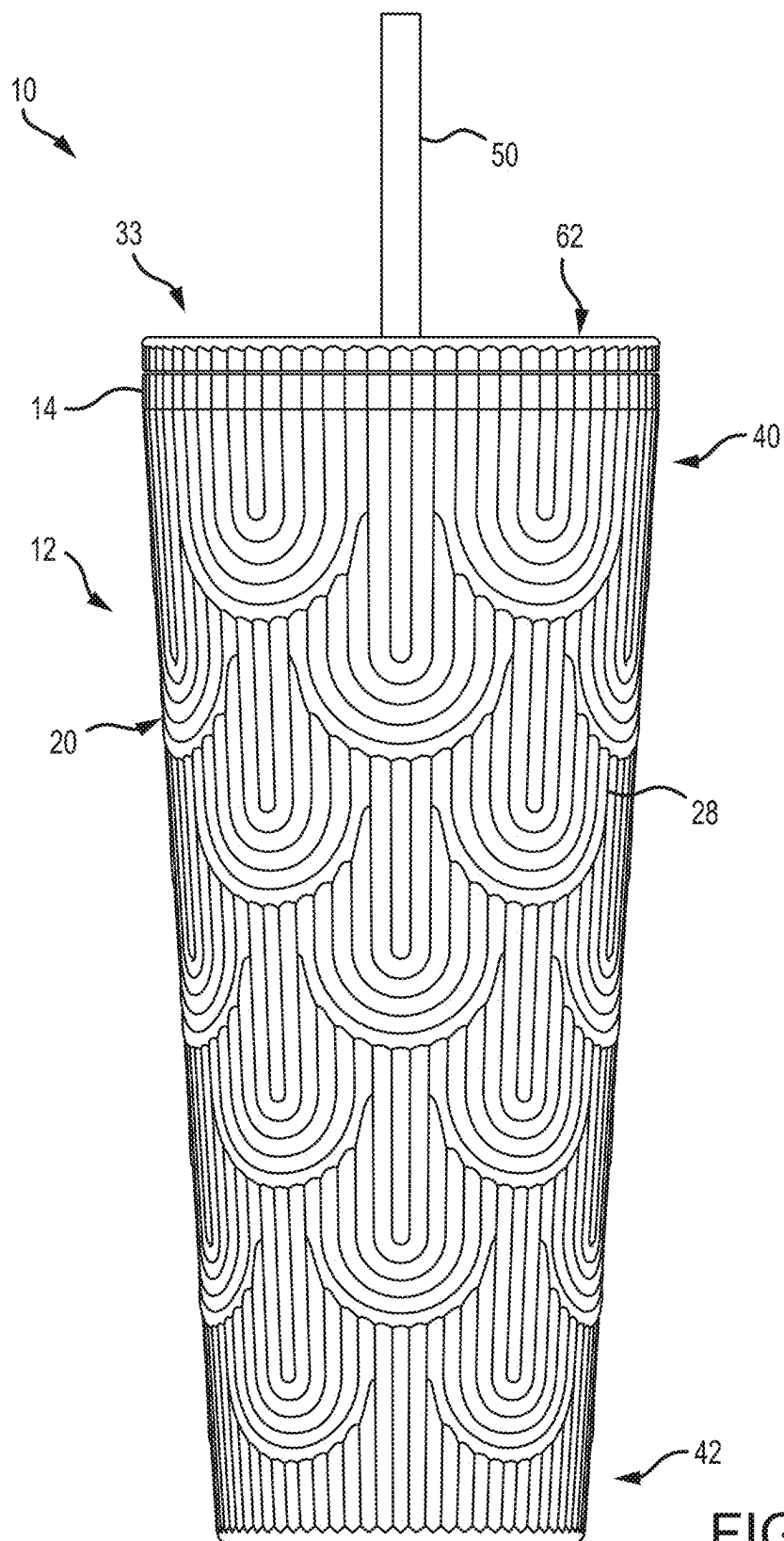


FIG.3

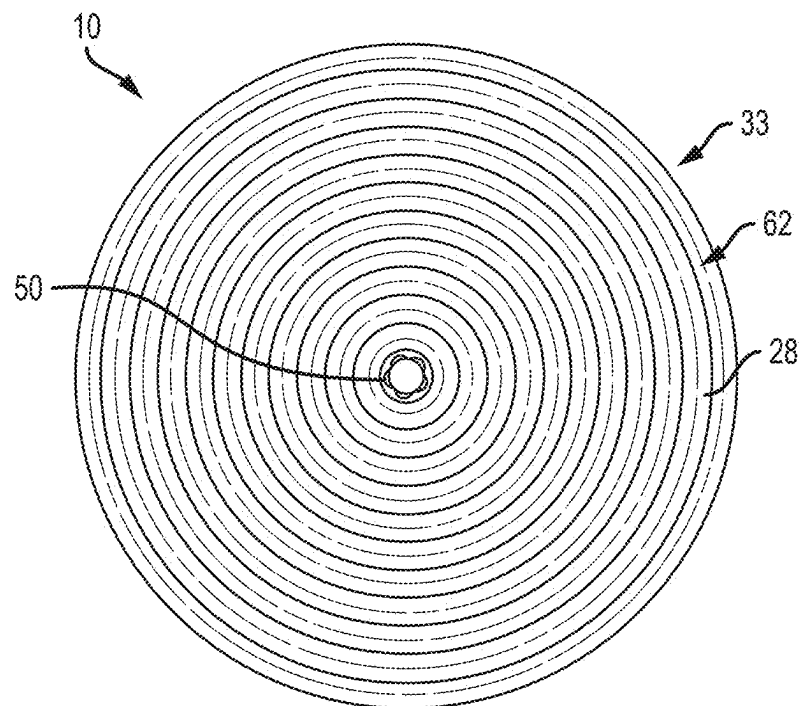


FIG.4

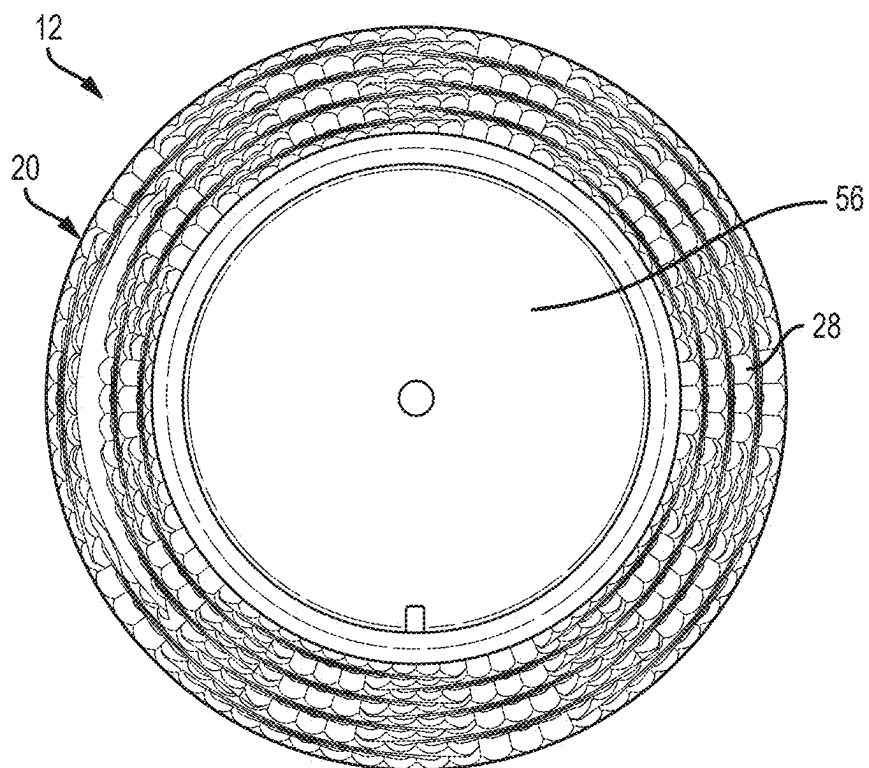


FIG.5

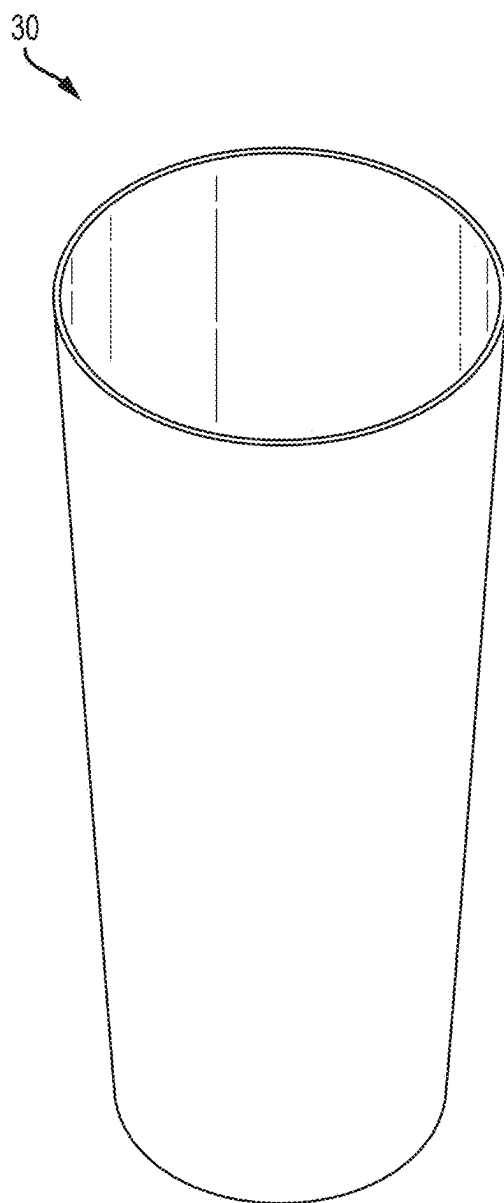


FIG.6

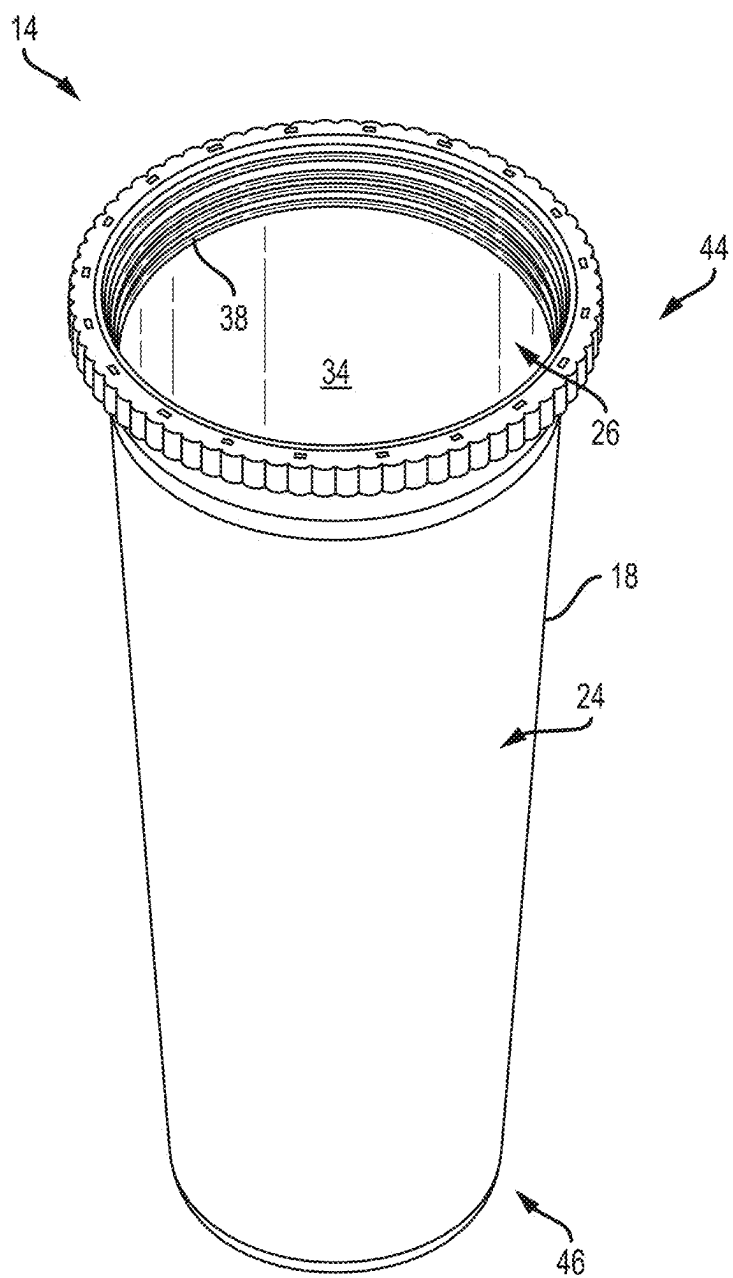
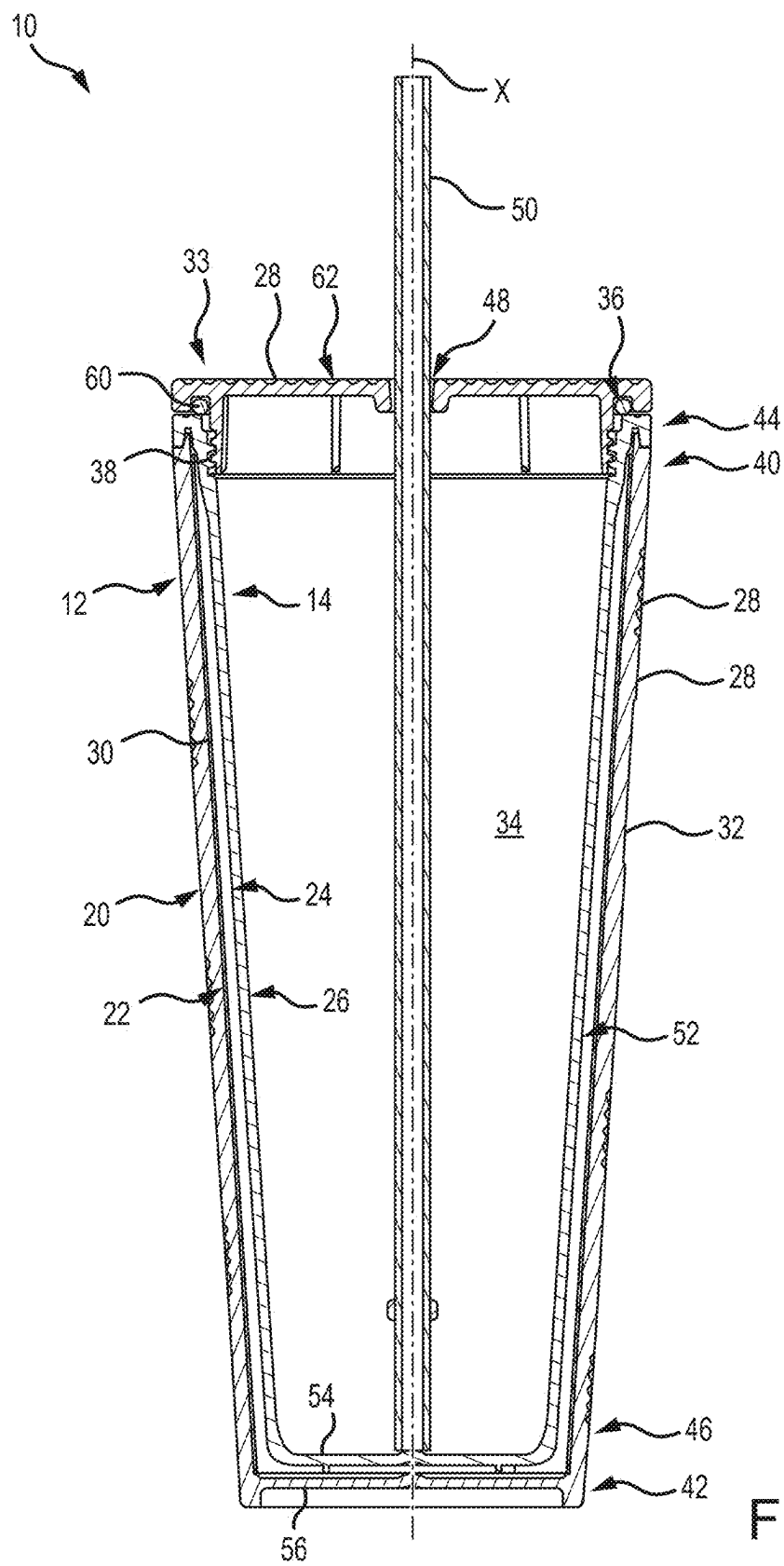


FIG. 7



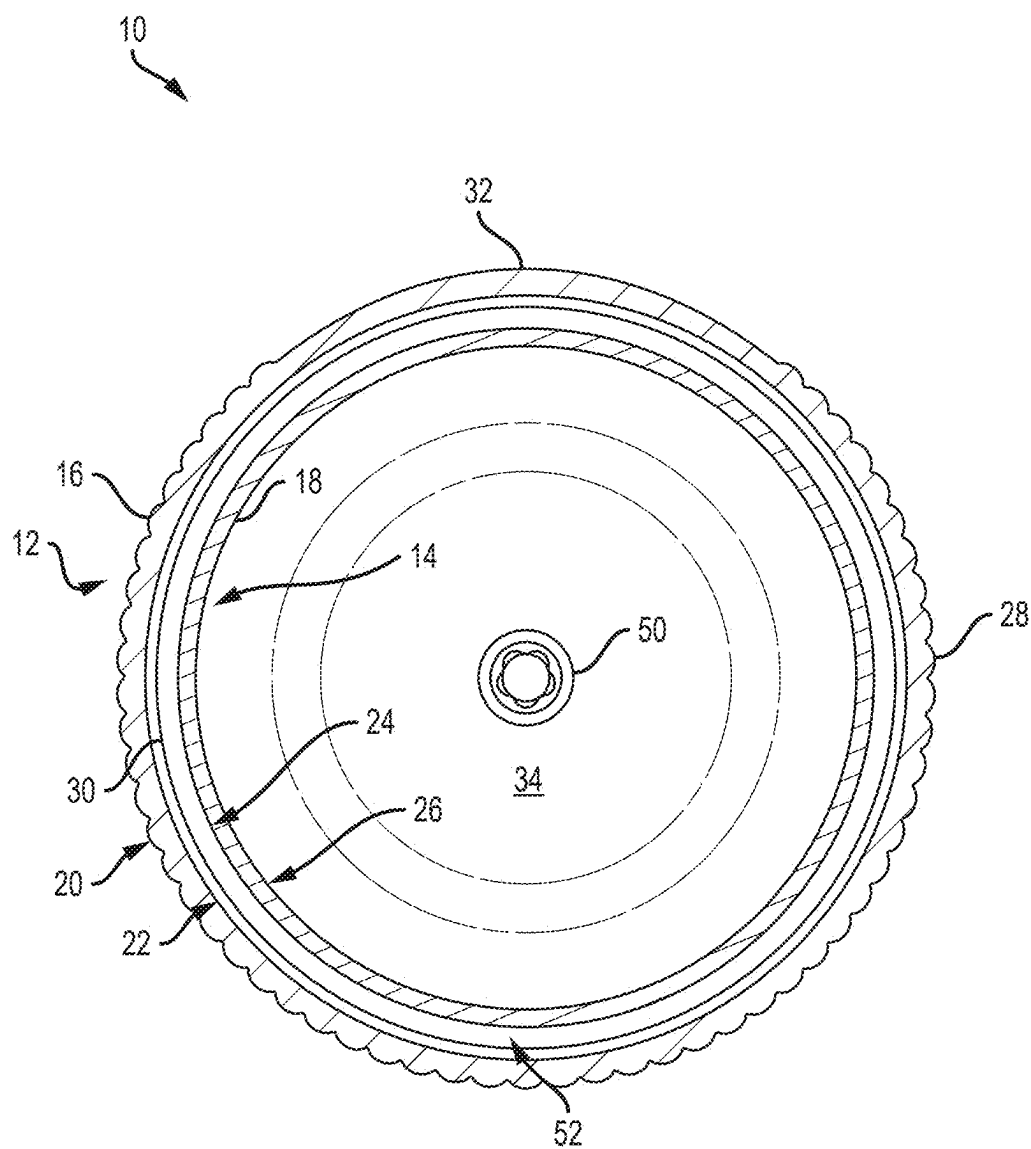


FIG.9

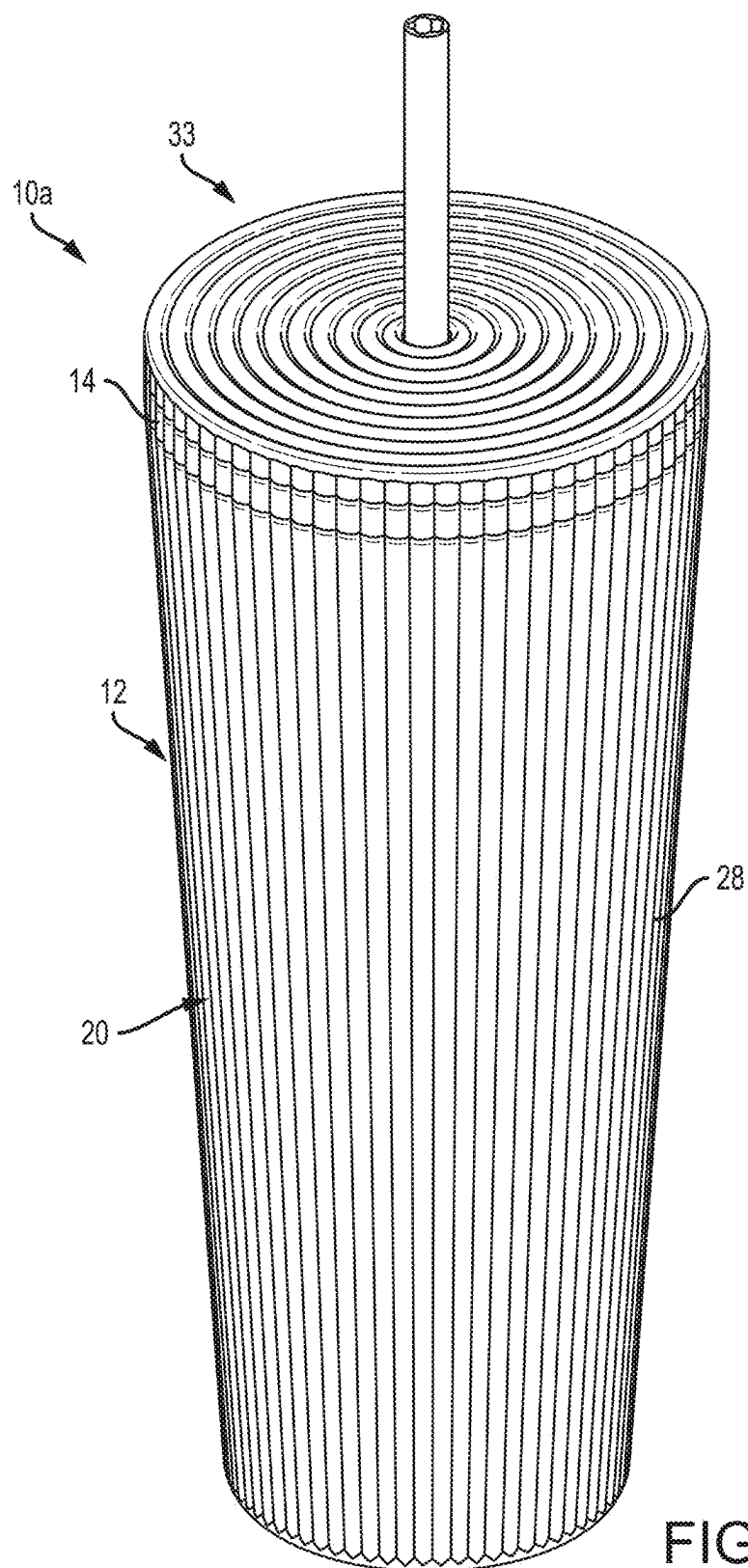
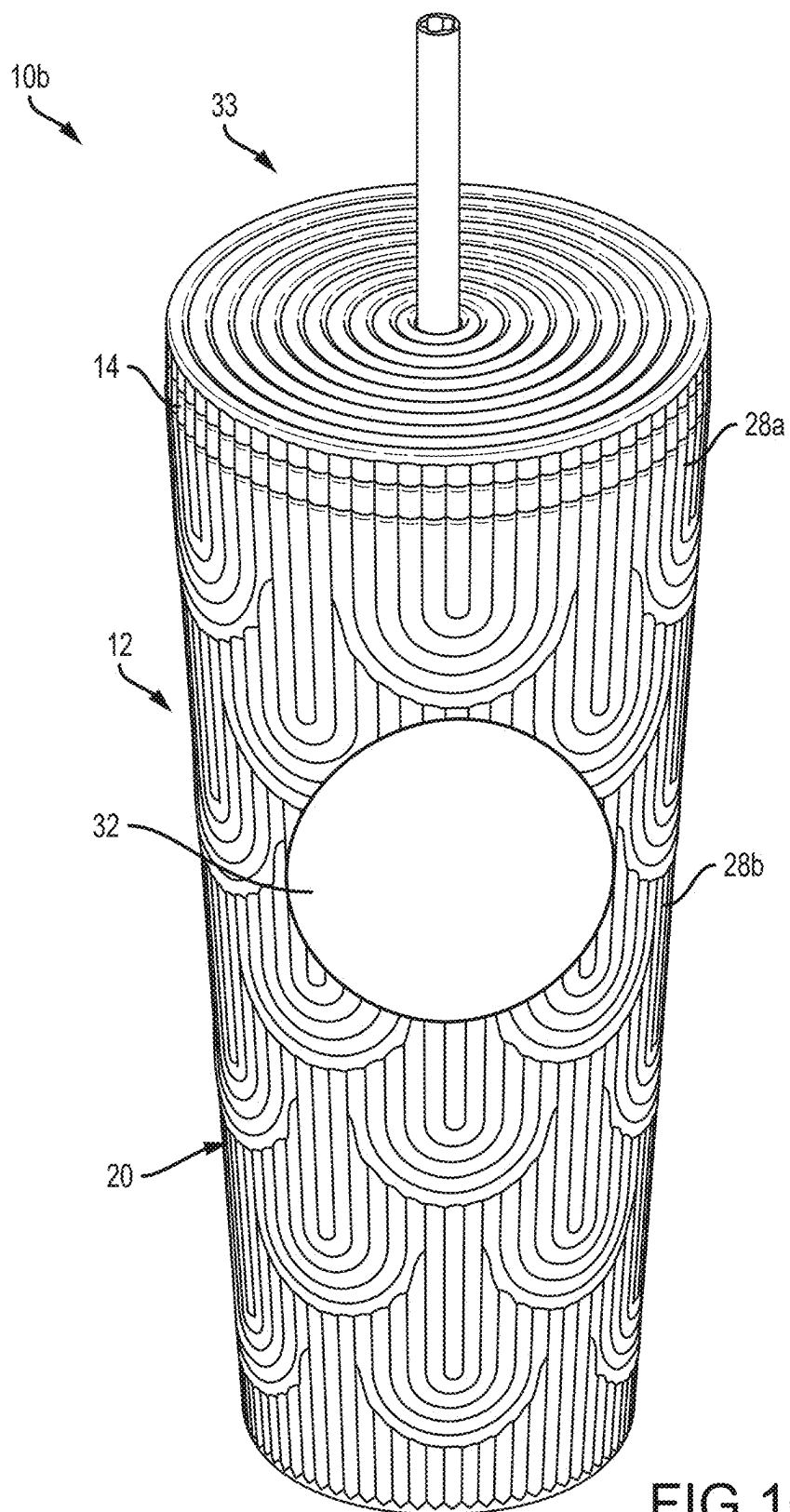


FIG. 10A



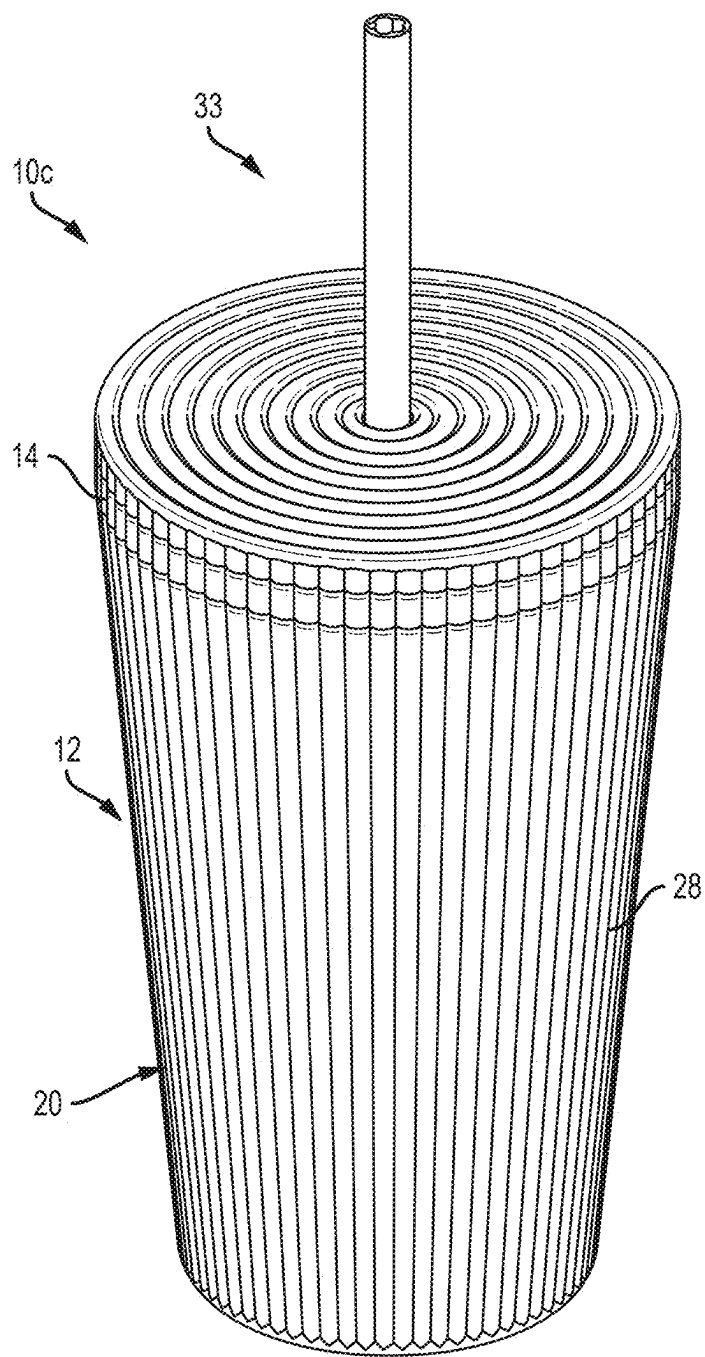
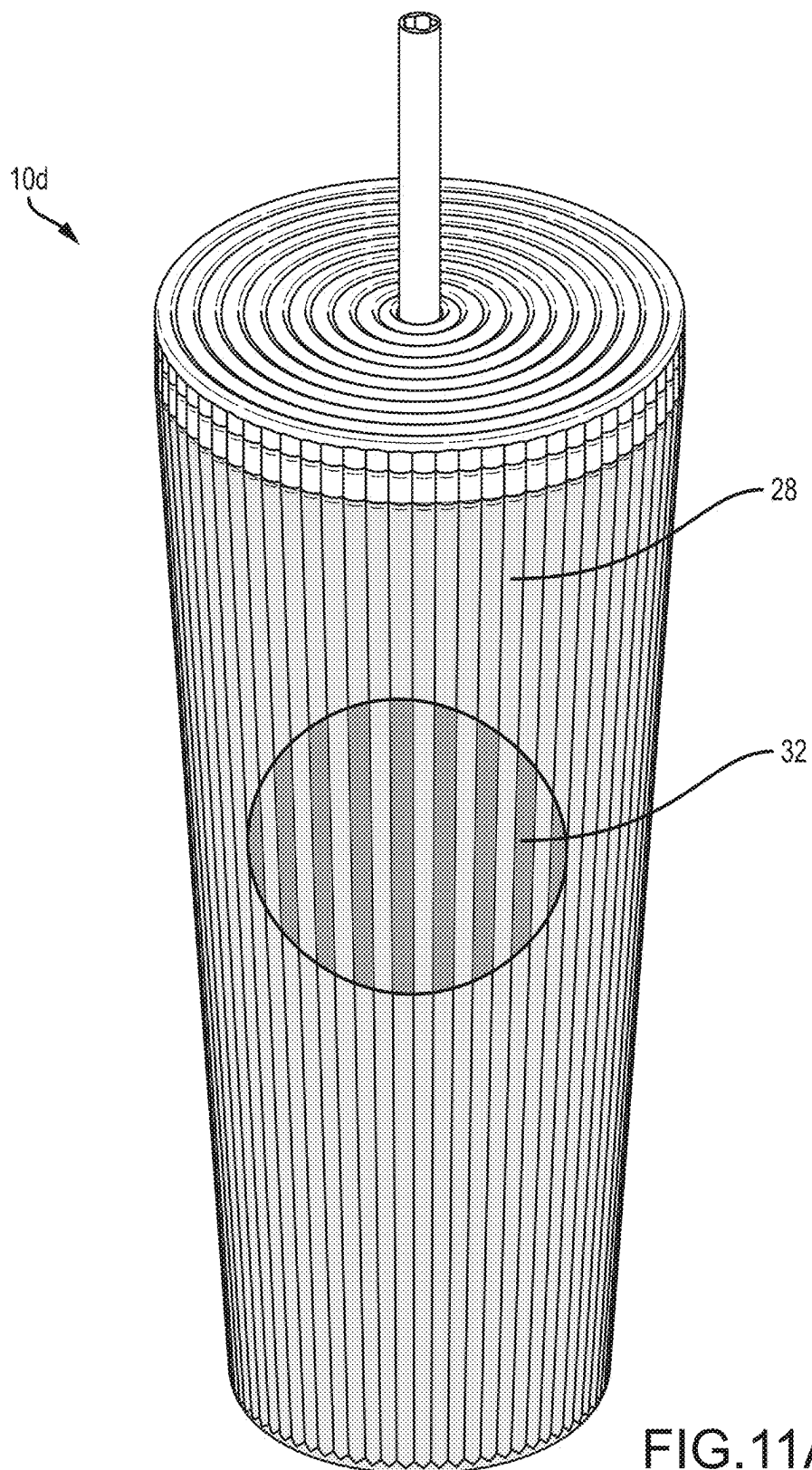
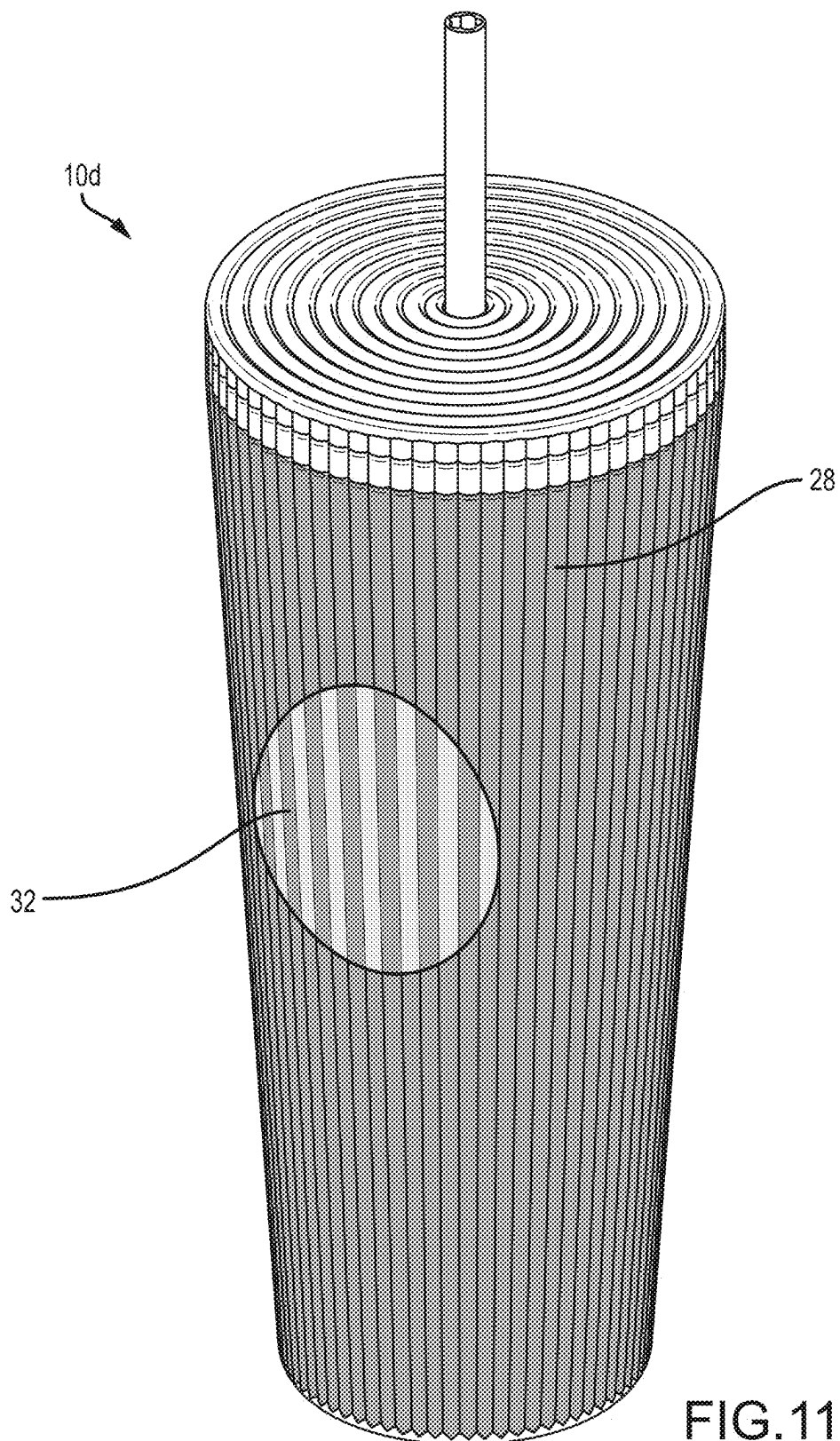
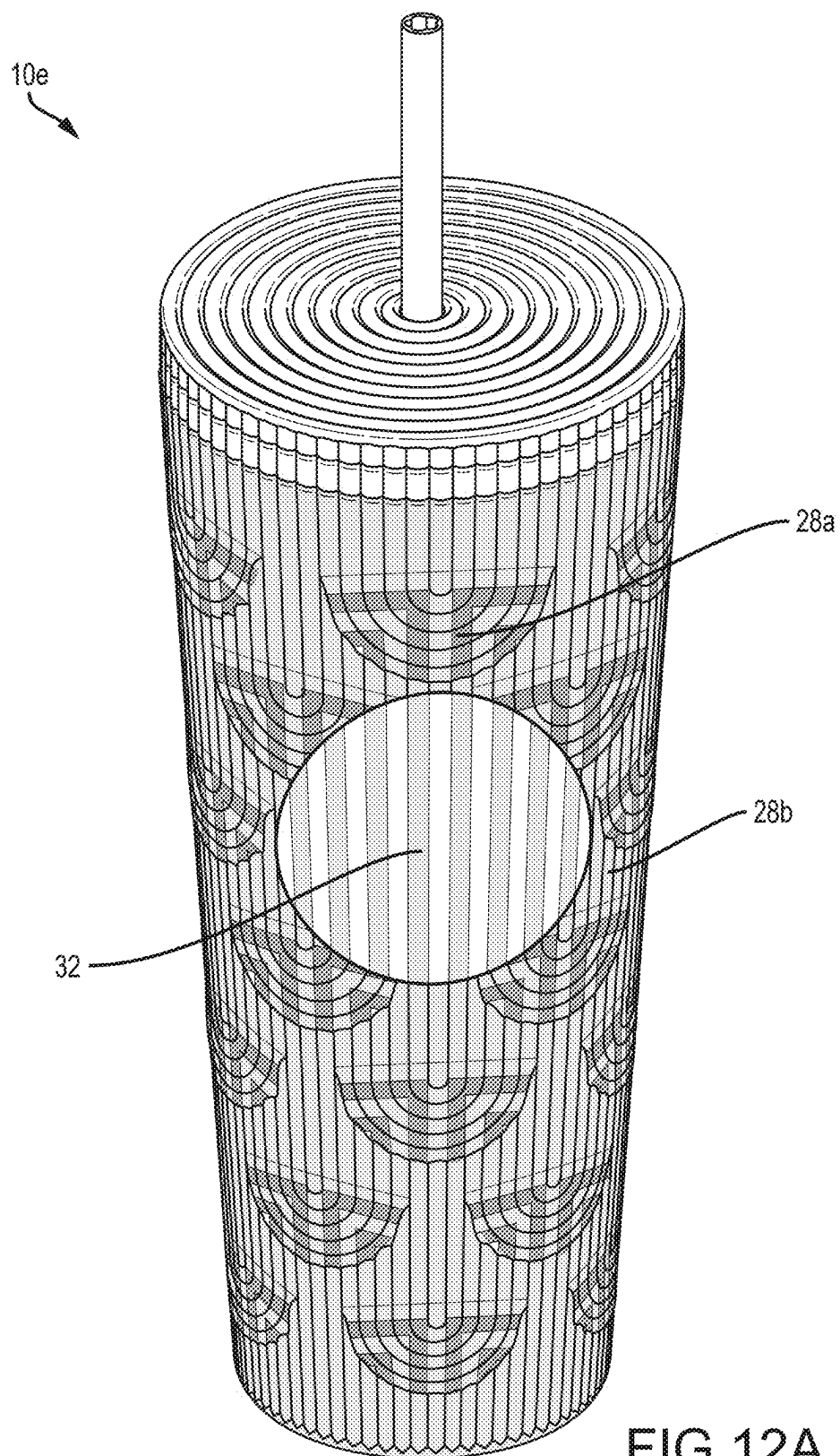
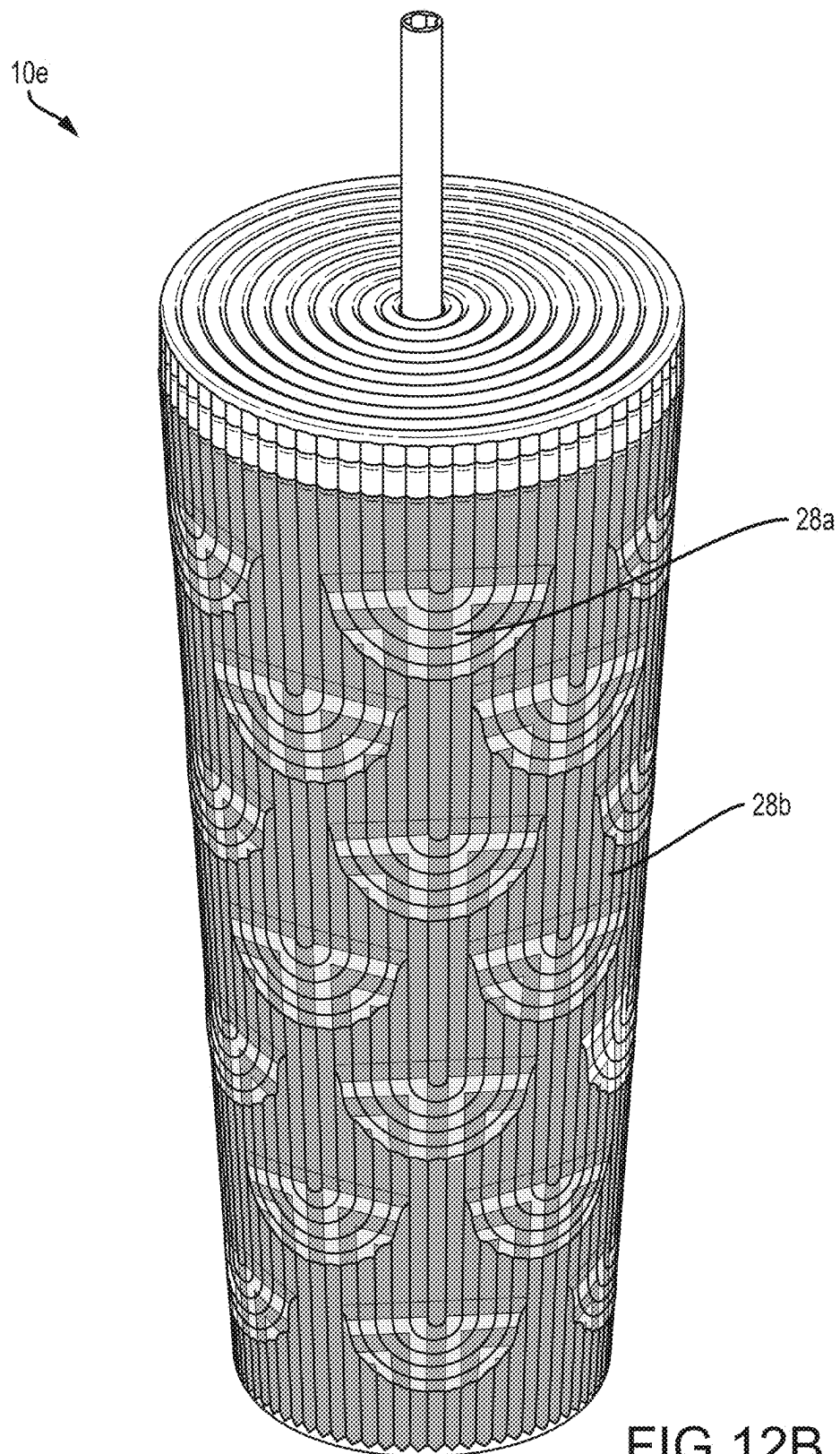


FIG. 10C









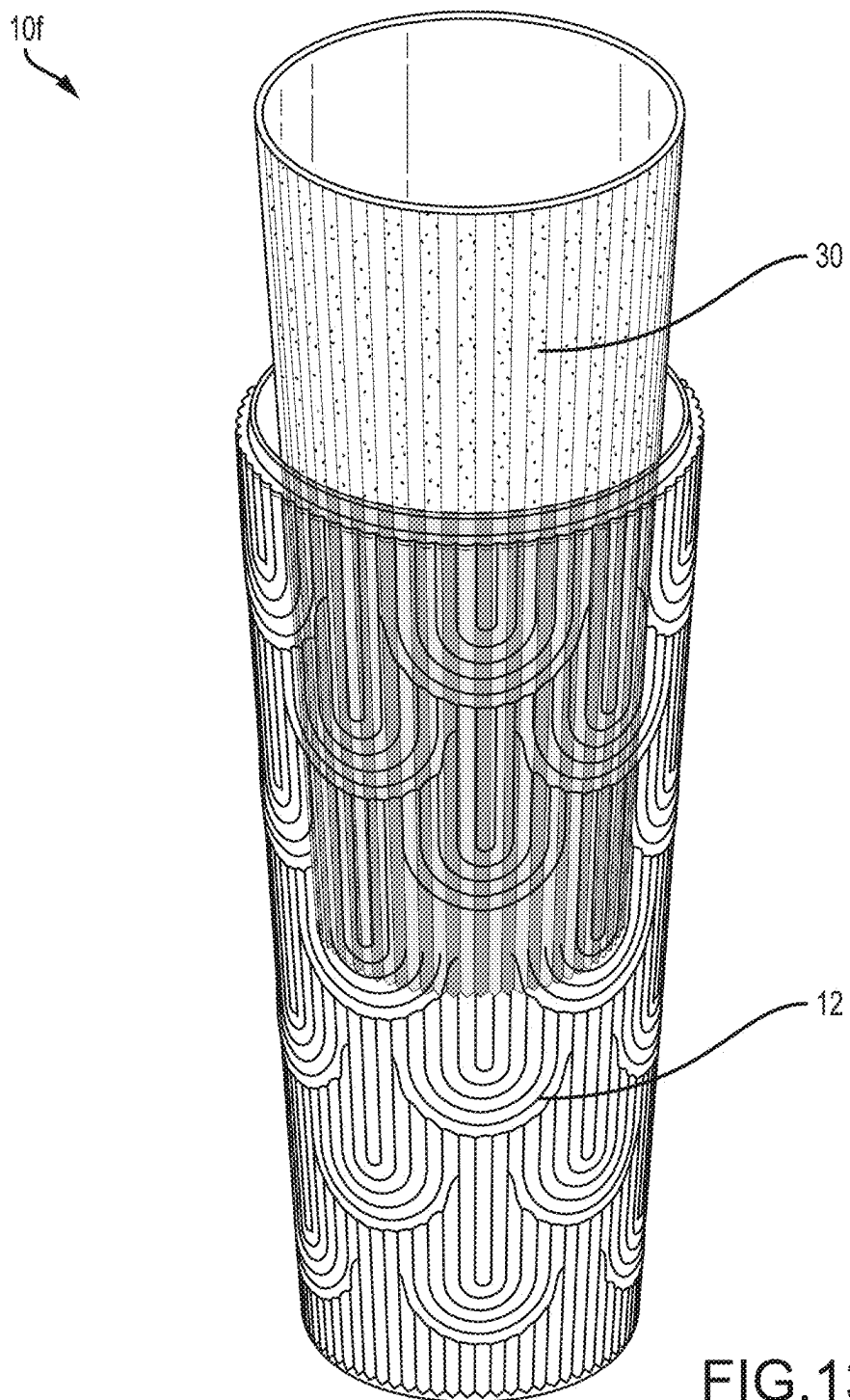


FIG.13

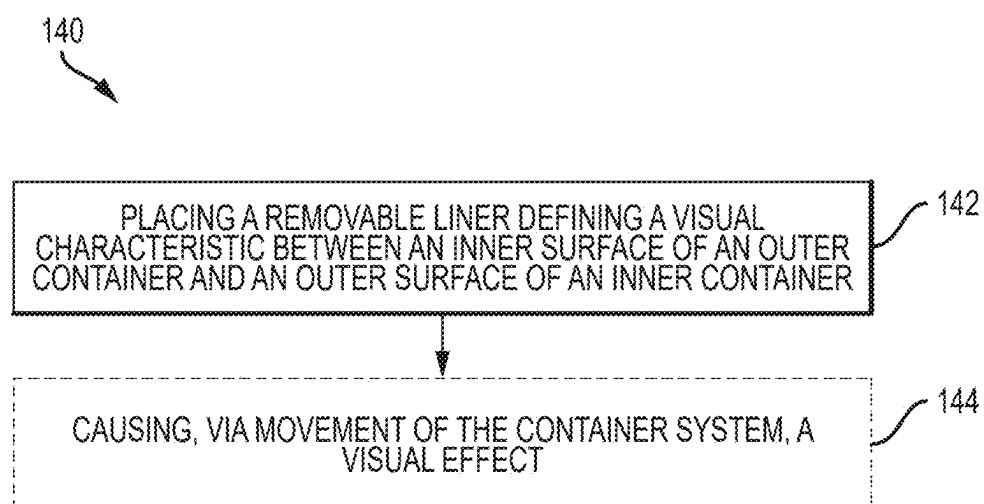


FIG.14

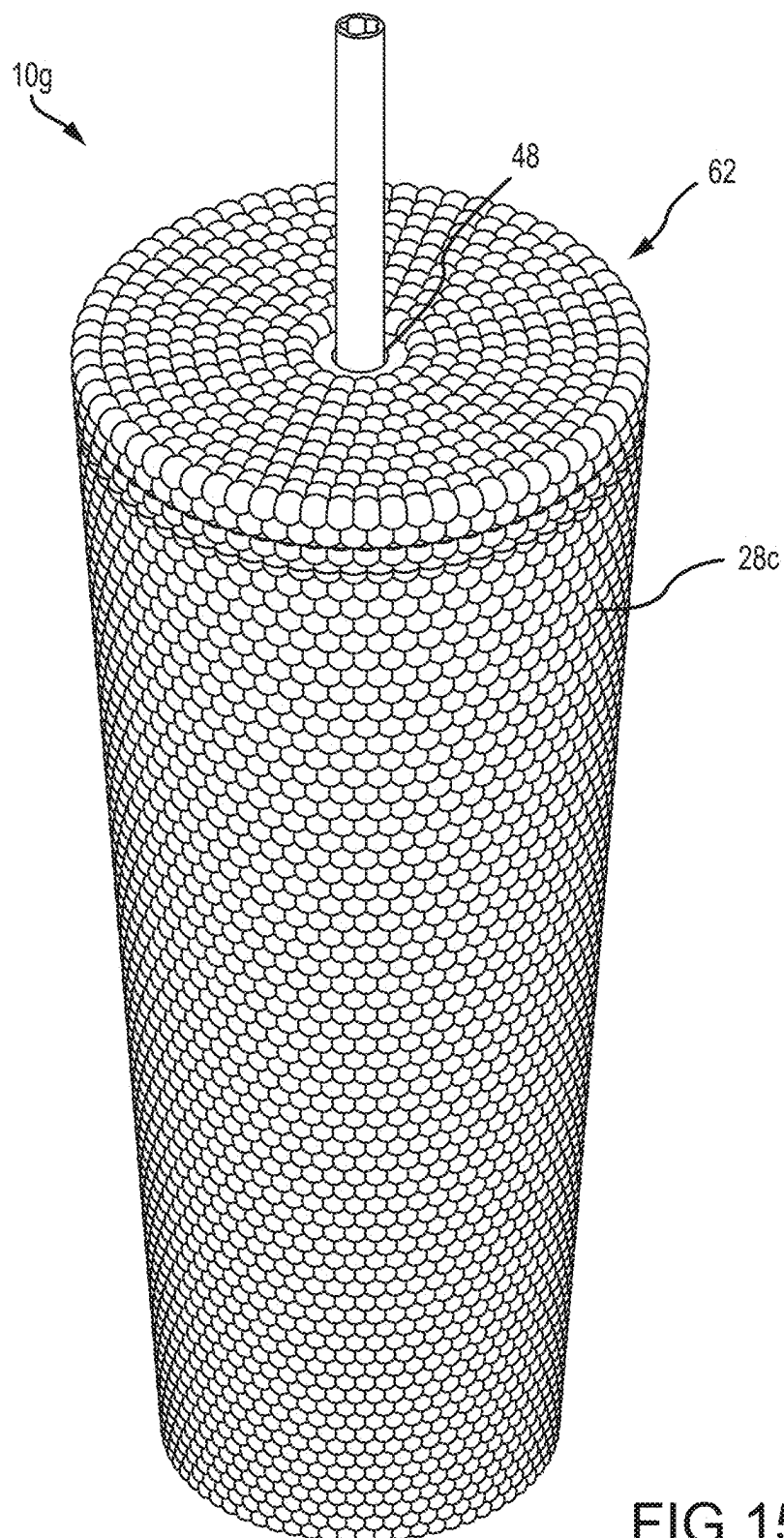


FIG. 15

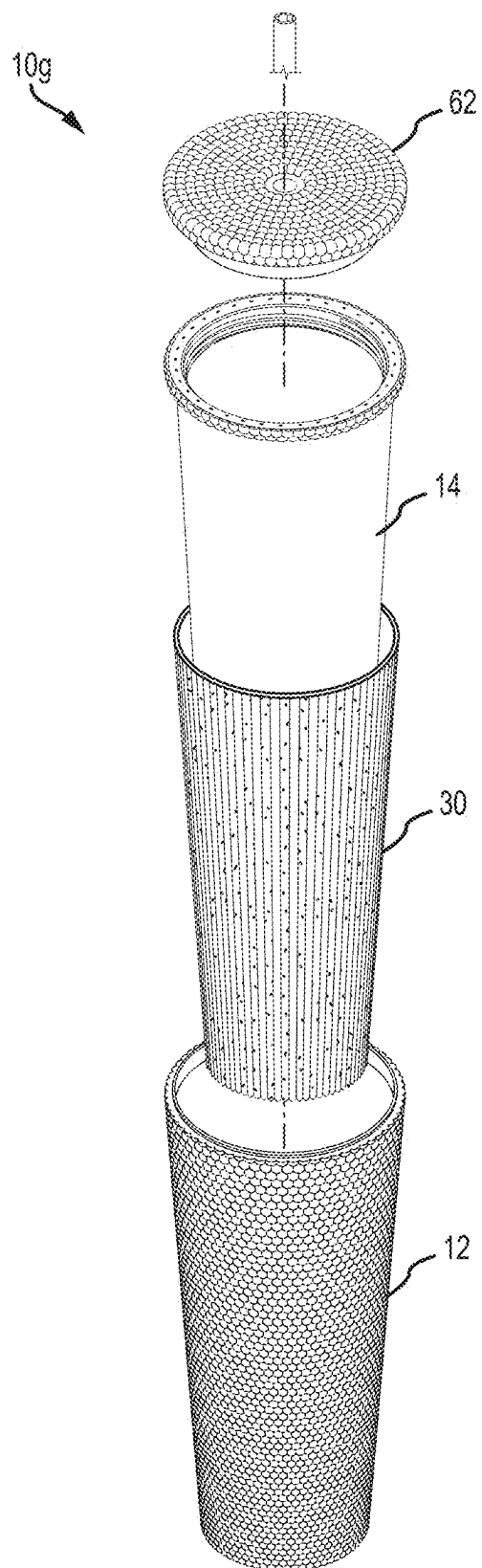


FIG.16

CONTAINER SYSTEM AND METHOD

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of U.S. Provisional Application Ser. No. 63/556,752, filed Feb. 22, 2024, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] In some examples, a container system that may have visual characteristics that may be altered by movement of the container system or movement of a viewpoint may be desirable.

SUMMARY

[0003] The present disclosure relates to a container system and method, as described herein, that includes one or more lenticular lenses on an exterior surface of the container system. The disclosed systems may provide a visual effect wherein a visual characteristic may change (for example, from a first color or design to a second color or design) when the container system is moved or viewed from a different angle. The color(s) and/or design(s) may be provided via an insert/liner that fits between the exterior layer of the container system (that includes the lenticular lens(es)) and an interior layer of the container system.

[0004] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Non-limiting and non-exhaustive examples are described with reference to the following Figures.

[0006] FIG. 1 illustrates a top perspective view of a container system, constructed in accordance with principles of this disclosure, according to an example.

[0007] FIG. 2 illustrates a bottom perspective view of the container system of FIG. 1, according to an example.

[0008] FIG. 3 illustrates a side view of the container system of FIG. 1, according to an example.

[0009] FIG. 4 illustrates a top view of the container system of FIG. 1, according to an example.

[0010] FIG. 5 illustrates a bottom view of the container system of FIG. 1, according to an example.

[0011] FIG. 6 illustrates a top perspective view of a liner of the container system of FIG. 1, according to an example.

[0012] FIG. 7 illustrates a top perspective view of an inner container of the container system of FIG. 1, according to an example.

[0013] FIG. 8 illustrates a side cross-sectional view of the container system of FIG. 1, according to an example.

[0014] FIG. 9 illustrates a top-down cross-sectional view of the container system of FIG. 1, according to an example.

[0015] FIG. 10A illustrates a top perspective view of an embodiment of the container system of FIG. 1, including lenticular lenses, according to an example.

[0016] FIG. 10B illustrates a top perspective view of an embodiment of the container system of FIG. 1, including

multiple shapes of lenticular lenses and a non-lenticular feature, according to an example.

[0017] FIG. 10C illustrates a top perspective view of an embodiment of the container system of FIG. 1, including lenticular lenses, with a smaller height-to-diameter ratio, according to an example.

[0018] FIG. 11A illustrates a view of an embodiment of the container system of FIG. 1 at a first position, displaying a first visual effect, according to an example.

[0019] FIG. 11B illustrates a view of an embodiment of the container system of FIG. 1 at a second position, displaying a second visual effect, according to an example.

[0020] FIG. 12A illustrates a view of an embodiment of the container system of FIG. 1 at a first position, displaying a first visual effect, according to an example.

[0021] FIG. 12B illustrates a view of an embodiment of the container system of FIG. 1 at a second position, displaying a second visual effect, according to an example.

[0022] FIG. 13 illustrates a view of an embodiment of the container system of FIG. 1, wherein the lid and inner container are removed, and the liner is partially removed from the outer container, according to an example.

[0023] FIG. 14 illustrates an example method for forming a container system, according to an example.

[0024] FIG. 15 illustrates a top perspective view of another example container system, constructed in accordance with principles of this disclosure.

[0025] FIG. 16 illustrates an exploded top perspective view of the container system of FIG. 15.

DETAILED DESCRIPTION

[0026] In the following detailed description, references are made to the accompanying drawings that form a part hereof, and in which are shown by way of illustrations specific embodiments or examples. These aspects may be combined, other aspects may be utilized, and structural changes may be made without departing from the present disclosure. Examples may be practiced as methods, systems, or devices. The following detailed description is therefore not to be taken in a limiting sense, and the scope of the present disclosure is defined by the appended claims and their equivalents.

[0027] In some examples, a user may desire a container system (such as a cup for holding, transporting, and drinking a beverage, for example) that has a particular design, image, or color. In some examples, the user may desire a container system that may have a visual characteristic that may change when the container system is moved or viewed from a different angle. In some examples, such a changeable visual characteristic may be achieved via the use of one or more lenticular lenses as part of the container system.

[0028] Disclosed herein are example embodiments of a container system that includes one or more lenticular lenses on an exterior surface of the container system. The disclosed container systems provide a visual characteristic that may appear different to the viewer/user when the container system is moved or viewed from a different angle. For example, when rotated or tilted with respect to the viewer, a container system that appears to be a first color originally may then appear to be a second color. In another example, when rotated or tilted with respect to the viewer, a container system that appears to have a first design (e.g., pattern, image, logo, art) may then appear to have a second design. The color(s) and/or design(s) may be provided via an

insert/liner that fits between the exterior layer of the container system (that includes the lenticular lens(es)) and an interior layer of the container system (that is configured to hold a liquid).

[0029] These and other examples will be explained in more detail below with respect to FIG. 1 to FIG. 16.

[0030] In some examples, a container system 10 includes an outer container 12 that defines/includes a closed bottom 56 (at an outer container bottom end 42) and an opening/open top 68 (at an outer container top end 40). The outer container 12 includes/defines an at least partially (in some examples, completely) a transparent wall 16 that extends between an outer surface 20 of the outer container 12 and an inner surface 22 of the outer container 12. The outer container 12 defines/includes one or more lenticular lenses 28 on at least the outer surface 20 of the outer container 12.

[0031] In some examples, the container system 10 includes an inner container 14 that is configured to hold liquids and/or solids (e.g., food and/or beverages) within a defined interior volume 34. The inner container 14 defines an inner surface 26 for holding liquids, an open top 66 (at an inner container top end 44), a closed bottom end 54 (at an inner container bottom end 46), and an outer surface 24. Inner container 14 includes/defines an inner container wall 18 that extends between the inner surface 26 and the outer surface 24. When inner container 14 is inserted into outer container 12, an inter-wall volume 52 is defined between the inner surface 22 of the outer container 12 and the outer surface 24 of the inner container 14. In some examples, inner container 14 and outer container 12 are removably connected. In some examples, inner container 14 and outer container 12 are permanently connected. In some examples, inner container 14 and outer container 12 are connected by pressure fit, friction fit, snaps, threads, other connection types, soldering, ultrasonic welding, may be formed/manufactured together, or other means of connection. In some examples, assembled container system 10 is defined along a vertical axis X.

[0032] In some examples, a liner 30 defining a visual characteristic is inserted into the inter-wall volume 52. The liner 30 may be flexible or rigid. The liner 30 may be non-removable, in some examples. In some examples, the liner 30 may be removable. The liner 30 may include a pattern or other layout of colors or other designs that allows for the visual characteristic to change when viewed through the lenticular lens(es) 28. When inserted into the inter-wall volume 52, the liner 30 is captured between the inner surface 22 of the outer container 12 and the outer surface 24 of the inner container 14. Movement of the container system 10 (or movement of the viewer in relation to the container system 10) causes a visual effect that can be seen from the outer surface 20 of the outer container 12.

[0033] In some examples, the lenticular lens(es) 28 create a textured surface on the outer surface 20 of the outer container 12. In some examples, a majority (or the entirety) of the outer surface 20 of the outer container 12 defines lenticular lenses 28. In some examples, a non-lenticular feature 32 is present on the outer surface 20, through which all or part of the liner may be viewed in an unaltered format. In some examples, a visual characteristic of the liner 30 is defined by a color. In some examples, the visual characteristic of the liner 30 is defined by an image, pattern, or other design. In some examples, the outer surface 20 of the outer container 12 includes lenticular lenses 28 that are oriented in

at least two different directions. In some examples, outer surface 20 of the outer container 12 includes lenticular lenses 28 that have a curved configuration. In some examples, outer container 12 includes a plurality of lenticular lenses 28 that are oriented parallel to each other. In some examples, the outer surface 20 of the outer container 12 includes a plurality of lenticular lenses 28, a majority of the lenses 28 extending from the top end 40 of the outer container to the bottom end 42 of the outer container. In some examples, lenses 28a may be interrupted by other lenses 28b that are in a different orientation. In some examples, lenses 28 may be interrupted by a non-lenticular feature.

[0034] In some examples, the inner container 14 and/or the outer container 12 is closable by a lid 33. In an example, lid 33 may also have lenticular lenses 28 on an outer/top surface 62 of the lid 33. In an example, the lid 33 includes a sealing surface 36 that seals against the inner container 14 and/or the outer container 12. In an example, the inner container 14 and/or the outer container 12 is configured to connect to a connection feature of the lid 33. In a particular example, an inner connection feature 38 (for example, a threaded connection) of the lid 33 engages with inner container 14. In an example, an outer connection feature of lid 33 may engage with an outer container 12. In some examples, a sealing member 60 (for example, a gasket) provides a liquid seal against the inner container 14 and/or the outer container 12. In some examples, lid 33 includes a lid opening 48 through which a liquid may be consumed, or through which a straw 50 may be inserted.

[0035] In a particular example illustrated, container system 10a may include a single type of lenticular lens 28 (for example, running parallel vertically, circumferentially, diagonally, or in another arrangement or consistent pattern). Refer to FIG. 10A. In a particular example illustrated, container system 10b may include a several types of lenticular lenses 28a and 28b (for example, lenses 28a running parallel to each other vertically, and 28b arranged in a curved shape). Example container system 10b also includes a non-lenticular feature 32 (which interrupts one or more lenses 28a, 28b and through which a liner may be viewed). Refer to FIG. 10B. In a particular example illustrated, container system 10c may include at least one lenticular lens 28 and may have an alternate height-to-diameter ratio. Refer to FIG. 10C.

[0036] In a particular example illustrated, container system 10d may include a single type of lenticular lens 28, and when oriented in (or viewed from) a first position, may display a first visual effect (color/design). Refer to FIG. 11A. In a particular example illustrated, container system 10d may, when oriented in (or viewed from) a second position, may display a second visual effect (color/design). Refer to FIG. 11B. Where a non-lenticular feature 32 is present, the view of the liner may not change between the first position and the second position.

[0037] In a particular example illustrated, container system 10e may include a several types of lenticular lenses 28a and 28b, and when oriented in (or viewed from) a first position, may display a first visual effect (color/design). The first visual effect may differ between lenticular lenses 28a and 28b. Refer to FIG. 12A. In a particular example illustrated, container system 10e may, when oriented in (or viewed from) a second position, may display a second visual

effect (color/design). The second visual effect may differ between lenticular lenses **28a** and **28b**. Refer to FIG. **12B**.

[0038] In a particular example, container system **10f** may be configured so that the inner container **14** can be removed from the outer container **12** (for example, by a user or during manufacture). Example container system **10f** may be configured so that the liner **30** is removable (for example, by a user or during manufacture) from the outer container **12**. Refer to FIG. **13**.

[0039] FIG. **14** illustrates an example method **140** for forming a container system **10** as described herein. At operation **142**, a (removable, in some examples) liner defining a visual characteristic is placed between an inner surface of an outer container, wherein the inner surface defines a closed bottom and an open top, and an outer surface of an inner container. The outer container defines an at least partially transparent wall extending between an outer surface of the outer container and the inner surface of the outer container. The outer container defines a lenticular lens on at least the outer surface of the outer container. The inner container is configured for holding liquids and defines an inner surface for holding liquids. Insertion of the inner container within the outer container captures the liner between the inner surface of the outer container and the outer surface of the inner container.

[0040] In an example, the inner container is friction-fit to the outer container to form the container system **10**. In an example, the inner container is ultrasonically welded to the outer container to form the container system **10**. In an example, the inner container and/or the outer container is closable by a lid.

[0041] In some examples, at operation **144**, movement of the container system **10** (and/or movement of a viewer in relation to container system **10**) causes a visual effect that can be seen from the outer surface of the outer container.

[0042] In an example, the lenticular lens creates a textured surface on the outer surface of the outer container. In an example, a majority of the outer surface of the outer container defines lenticular lenses. In an example, the visual characteristic of the liner is defined by a color. In an example, the visual characteristic of the liner is defined by an image, design, or illustration. In an example, the outer surface of the outer container includes lenticular lenses that are oriented in at least two different directions. In an example, the outer surface of the outer container includes lenticular lenses that have a curved configuration. In an example, the outer surface of the outer container includes a plurality of lenticular lenses that are oriented parallel to each other. In an example, the outer surface of the outer container includes a non-lenticular feature that interrupts the lenticular lens at at least one location. In an example, the outer surface of the outer container includes a plurality of lenticular lenses, and a majority of the lenses extend from a top end of the outer container to a bottom end of the outer container.

[0043] In accordance with principles of this disclosure, materials of construction for the container system **10** and its components as described herein include materials which are compatible with: the environment, beverages, and cleaning and sanitizing products. For example, the components of container system **10** may comprise plastic or other polymers, glass, ceramic, metal, other materials, and/or combinations of materials.

[0044] FIGS. **15** and **16** illustrate another example container system **10g** that includes another example lenticular

lens **28c**. In certain examples, the entirety, or near entirety, of the outer surface of the container system **10g** includes lenticular lenses **28c**. In certain examples, the lenticular lenses **28c** are dome shaped. In certain examples, the lenticular lenses **28c** are spherical or partly spherical. In certain examples, the lenticular lenses **28c** form alternating rows. In certain examples, a first row of lenticular lenses **28c** forms a circumferential ring around the container system **10g**. In certain examples, a pattern is formed by a second row of lenticular lenses **28c** where one of the lenticular lenses of the second row is positioned in a space defined by two neighboring lenticular lenses of the first row. In certain examples, the pattern formed by the first and second rows is repeated along the length of the container system **10g**. In certain examples, the lid **33** has the same lenticular lenses **28c**. In certain examples, the top surface **62** has the same pattern of rows as the outer container **12**, where the rows form rings around the lid opening **48**.

[0045] For the purposes of this application, terms such as “upper,” “lower,” “upward,” and “downward” are intended to be descriptive with reference to and in relation to the orientation shown in the Figures for clarity, but the examples as practiced and included in the scope of the claims may include examples where the systems and devices are in a different orientation.

[0046] While particular uses of the technology have been illustrated and discussed above, the disclosed technology can be used with a variety of environments in accordance with many examples of the technology. The above discussion is not meant to suggest that the disclosed technology is only suitable for implementation within the environments shown and described above.

[0047] This disclosure described some aspects of the present technology with reference to the accompanying drawings, in which only some of the possible aspects were shown. Other aspects can, however, be embodied in many different forms and should not be construed as limited to the aspects set forth herein. Rather, these aspects were provided so that this disclosure was thorough and complete and fully conveyed the scope of the possible aspects to those skilled in the art.

[0048] As should be appreciated, the various aspects described with respect to the figures herein are not intended to limit the technology to the particular aspects described. Accordingly, additional configurations can be used to practice the technology herein and/or some aspects described can be excluded without departing from the methods and systems disclosed herein.

[0049] Similarly, where operations of a process are disclosed, those operations are described for purposes of illustrating the present technology and are not intended to limit the disclosure to a particular sequence of operations. For example, the operations can be performed in differing order, two or more operations can be performed concurrently, additional operations can be performed, and disclosed operations can be excluded without departing from the present disclosure. Further, each operation can be accomplished via one or more sub-operations. The disclosed processes can be repeated.

[0050] Although specific aspects were described herein, the scope of the technology is not limited to those specific aspects. One skilled in the art will recognize other aspects or improvements that are within the scope of the present technology. Therefore, the specific structure, acts, or opera-

tions are disclosed only as illustrative aspects. The scope of the technology is defined by the following claims and any equivalents therein.

What is claimed is:

1. A container system comprising:
 - an outer container defining a closed bottom and an open top, the outer container defining an at least partially transparent wall extending between an outer surface of the outer container and an inner surface of the outer container, the outer container defining a lenticular lens on at least the outer surface of the outer container;
 - a liner defining a visual characteristic; and
 - an inner container for holding liquids, the inner container closable by a lid, wherein the inner container defines an inner surface for holding liquids and an outer surface, wherein insertion of the inner container within the outer container captures the liner between the inner surface of the outer container and the outer surface of the inner container, wherein movement of the container system causes a visual effect that can be seen from the outer surface of the outer container.
2. The container system of claim 1, wherein the lenticular lens creates a textured surface on the outer surface of the outer container.
3. The container system of claim 1, wherein a majority of the outer surface of the outer container defines lenticular lenses.
4. The container system of claim 1, wherein the visual characteristic of the liner is defined by a color.
5. The container system of claim 1, wherein the visual characteristic of the liner is defined by an image.
6. The container system of claim 1, wherein the inner container is friction-fit to the outer container to form the container system.
7. The container system of claim 1, wherein the outer surface of the outer container includes lenticular lenses that are oriented in at least two different directions.
8. The container system of claim 1, wherein the outer surface of the outer container includes a plurality of lenticular lenses that have a curved configuration.
9. The container system of claim 8, wherein the plurality of lenticular lenses are dome shaped.
10. The container system of claim 1, wherein the outer surface of the outer container includes a plurality of lenticular lenses that are oriented parallel to each other.
11. The container system of claim 1, wherein the outer surface of the outer container includes a plurality of lenticular lenses, wherein a majority of the lenses extend from a top end of the outer container to a bottom end of the outer container.
12. The container system of claim 1, wherein the liner is removable.
13. A method of forming a container system comprising:
 - placing a liner defining a visual characteristic between an inner surface of an outer container, wherein the inner surface defines a closed bottom and an open top, and an outer surface of an inner container, wherein:
 - the outer container defines an at least partially transparent wall extending between an outer surface of the outer container and the inner surface of the outer container,
 - the outer container defines a lenticular lens on at least the outer surface of the outer container,
 - the inner container is configured for holding liquids,
 - the inner container defines an inner surface for holding liquids, and
 - insertion of the inner container within the outer container captures the liner between the inner surface of the outer container and the outer surface of the inner container.
14. The method of claim 13, wherein movement of the container system causes a visual effect that can be seen from the outer surface of the outer container.
15. The method of claim 13, wherein the lenticular lens creates a textured surface on the outer surface of the outer container.
16. The method of claim 13, wherein a majority of the outer surface of the outer container defines lenticular lenses.
17. The method of claim 13, wherein the visual characteristic of the liner is defined by a color.
18. The method of claim 13, wherein the visual characteristic of the liner is defined by an image.
19. The method of claim 13, wherein the inner container is friction-fit to the outer container to form the container system.
20. The method of claim 13, wherein the outer surface of the outer container includes lenticular lenses that are oriented in at least two different directions.
21. The method of claim 13, wherein the outer surface of the outer container includes lenticular lenses that have a curved configuration.
22. The method of claim 13, wherein the outer surface of the outer container includes a plurality of lenticular lenses that are oriented parallel to each other.
23. The method of claim 13, wherein the outer surface of the outer container includes a plurality of lenticular lenses, wherein a majority of the lenses extend from a top end of the outer container to a bottom end of the outer container.
24. The method of claim 13, wherein the inner container is closable by a lid.

* * * * *